

Lab: Identifying Insufficient Logging and Monitoring Failures

Objective: Identifying insufficient logging leading to monitoring failures

Tools required: Kali Linux

Prerequisites: Basic understanding of system and Apache log files

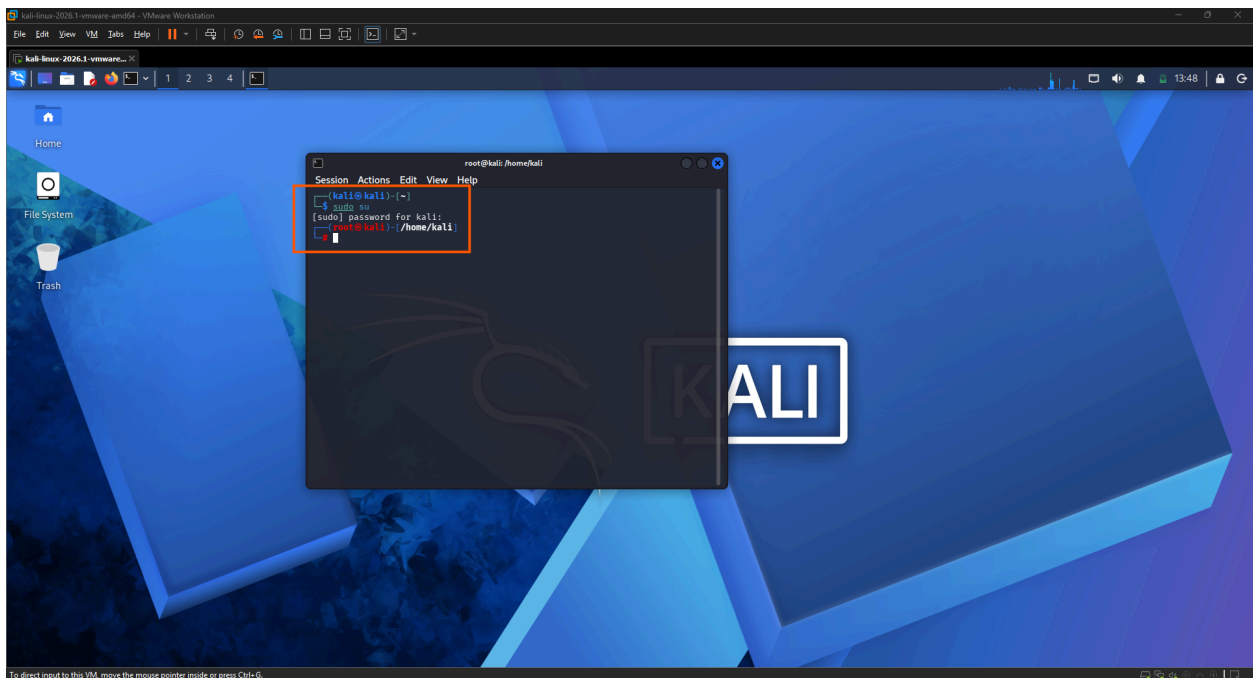
Steps to be followed:

1. Misconfiguring a web server for no logging
2. Adding the entries to log the connections happening to Apache server

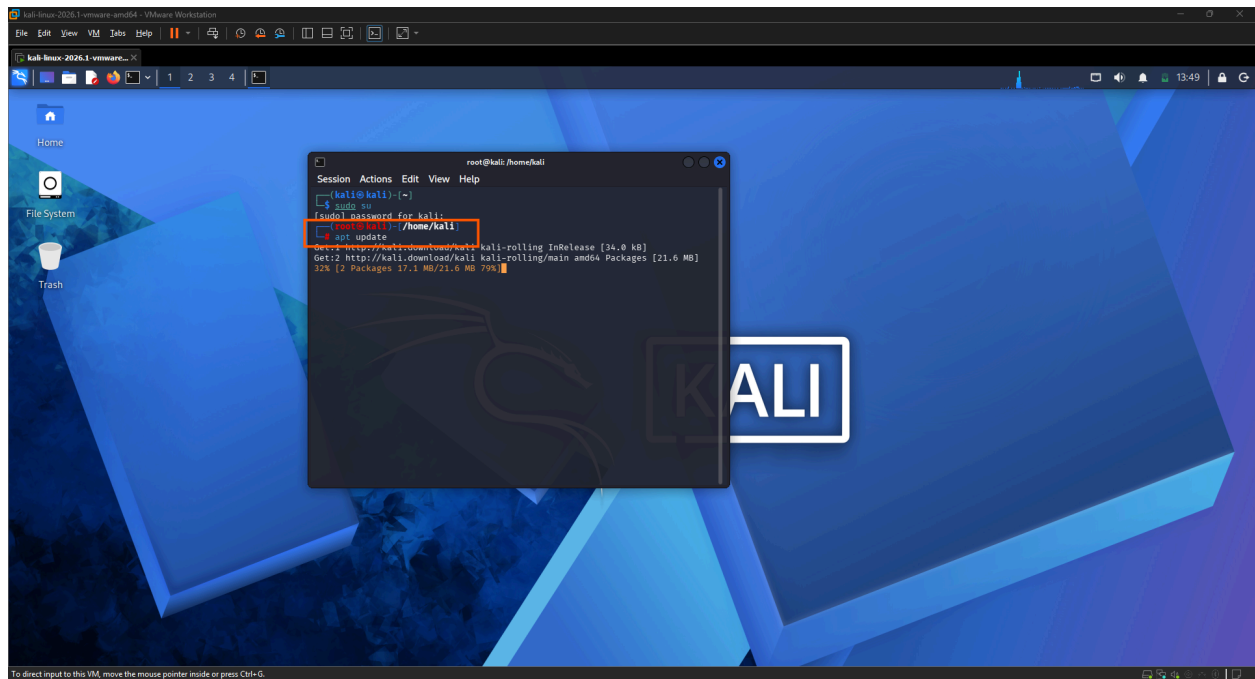
Step 1: Misconfiguring a web server for no logging

1.1 Launch Kali Linux VM, Open Terminal & Give suduers permission using the command:

sudo su

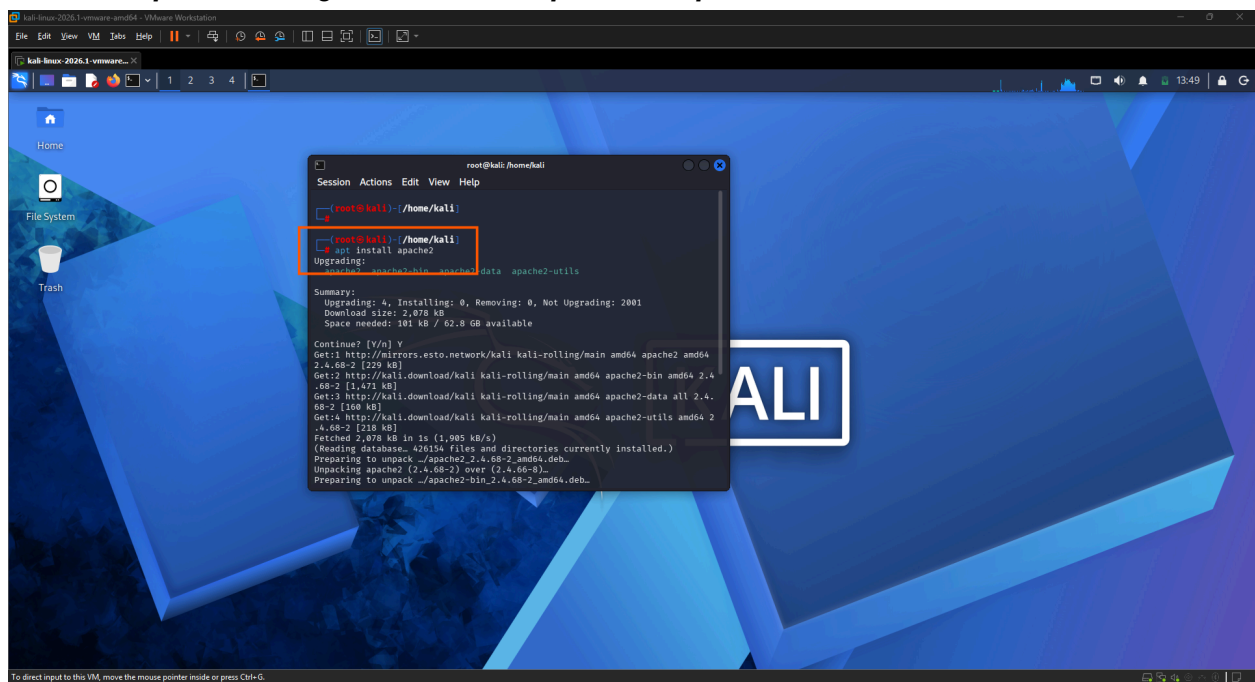


1.2 Use the command: ***apt update***



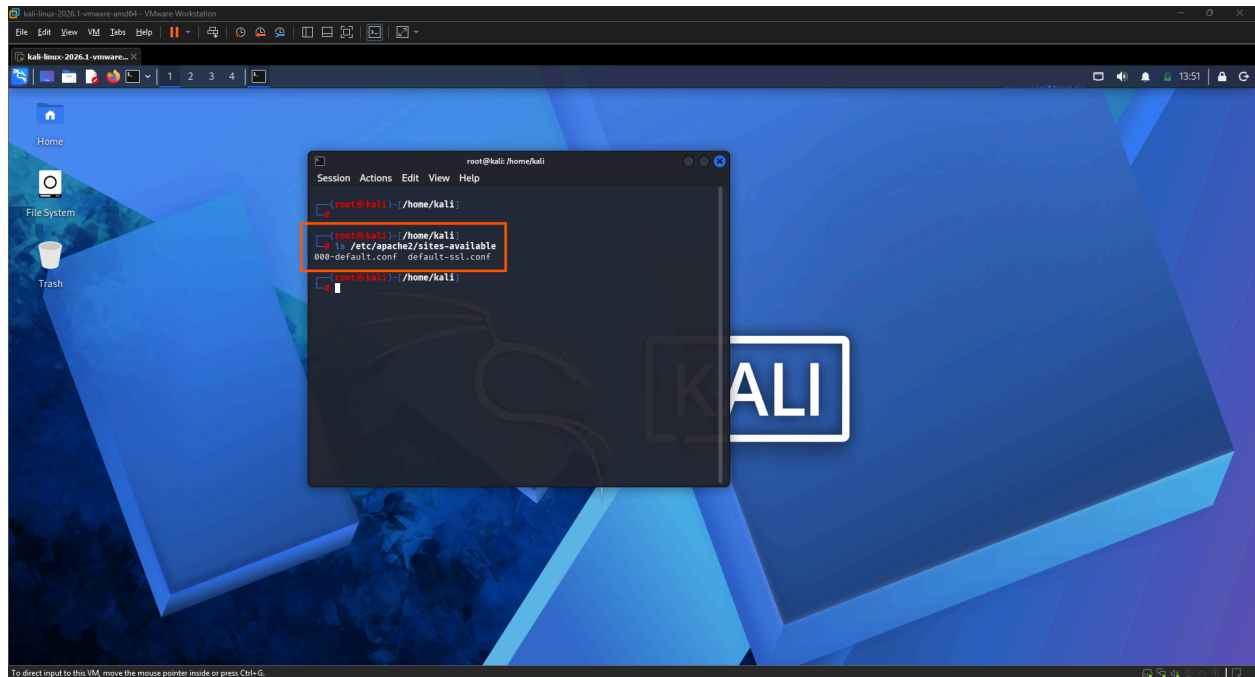
```
root@kali: /home/kali
(kali@kali)~$ sudo apt update
[sudo] password for kali:
(kali@kali)~$ apt update
Get:1 http://mirrors.esto.network/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.6 MB]
32% [2 Packages 17.1 MB/21.6 MB 79%]
```

1.3 Install ***apache2*** using the command: ***apt install apache2***

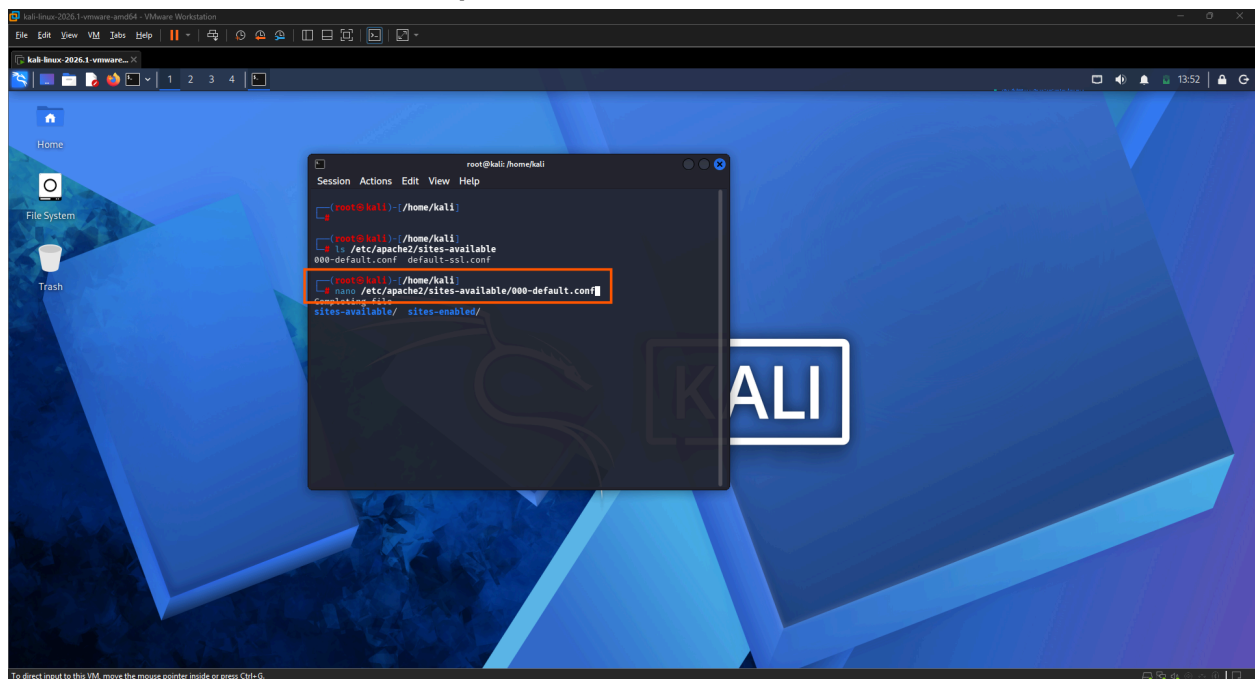


```
root@kali: /home/kali
(kali@kali)~$ apt install apache2
Upgrading:
Package apache2-bin apache2-data apache2-utils
Summary:
Upgrading: 4, Installing: 0, Removing: 0, Not Upgrading: 2001
Download size: 2,078 kB
Space needed: 101 kB / 62.8 GB available
Continue? [Y/n] Y
Get:1 http://mirrors.esto.network/kali kali-rolling/main amd64 apache2 amd64 2.4.68-2 [229 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 apache2-bin amd64 2.4.68-2 [1,472 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 apache2-data all 2.4.68-2 [180 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 apache2-utils amd64 2.4.68-2 [218 kB]
Fetched 2,078 kB in 1s (1,985 kB/s)
(Reading database, 42014 files and directories currently installed.)
Preparing to unpack ./apache2_2.4.68-2_amd64.deb...
Unpacking apache2 (2.4.68-2) over (2.4.66-8)...
Preparing to unpack ./apache2-bin_2.4.68-2_amd64.deb...
```

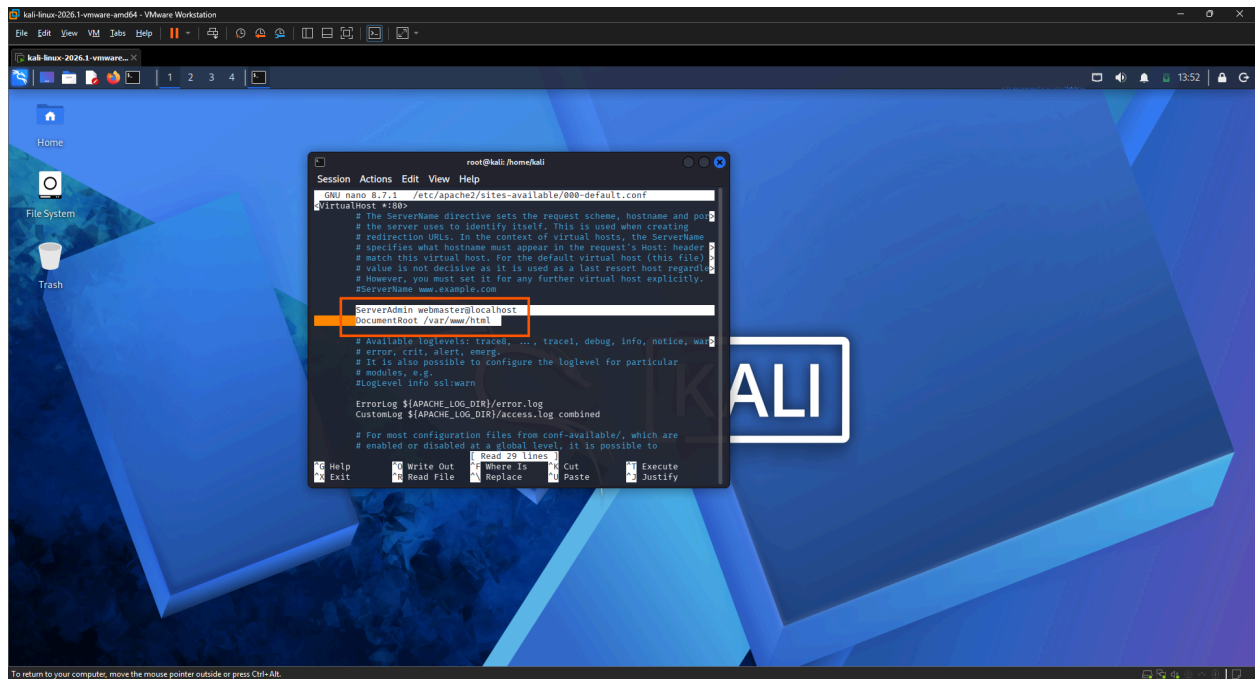
1.4 Now see the content inside **sites-available**, use the command:
ls /etc/apache2/sites-available



1.5 Use the command: **nano /etc/apache2/sites-available/000-default.conf**



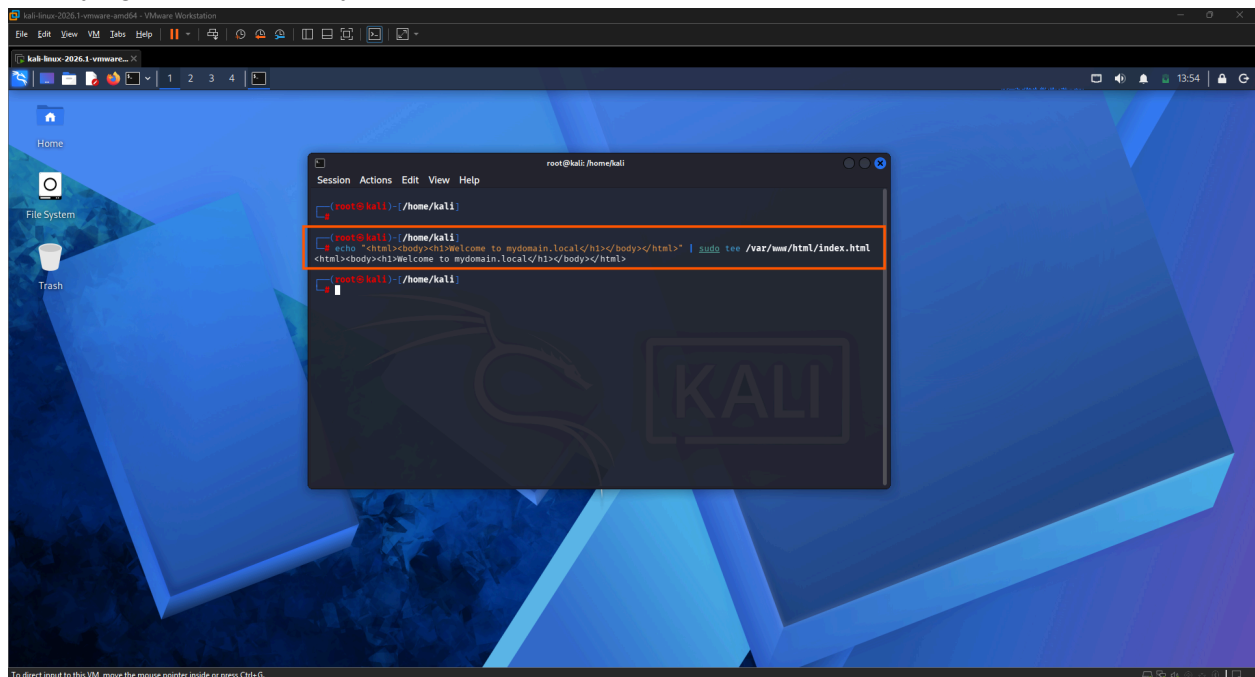
1.6 Remove the highlighted content & save the file.



1.7 Now use the command:

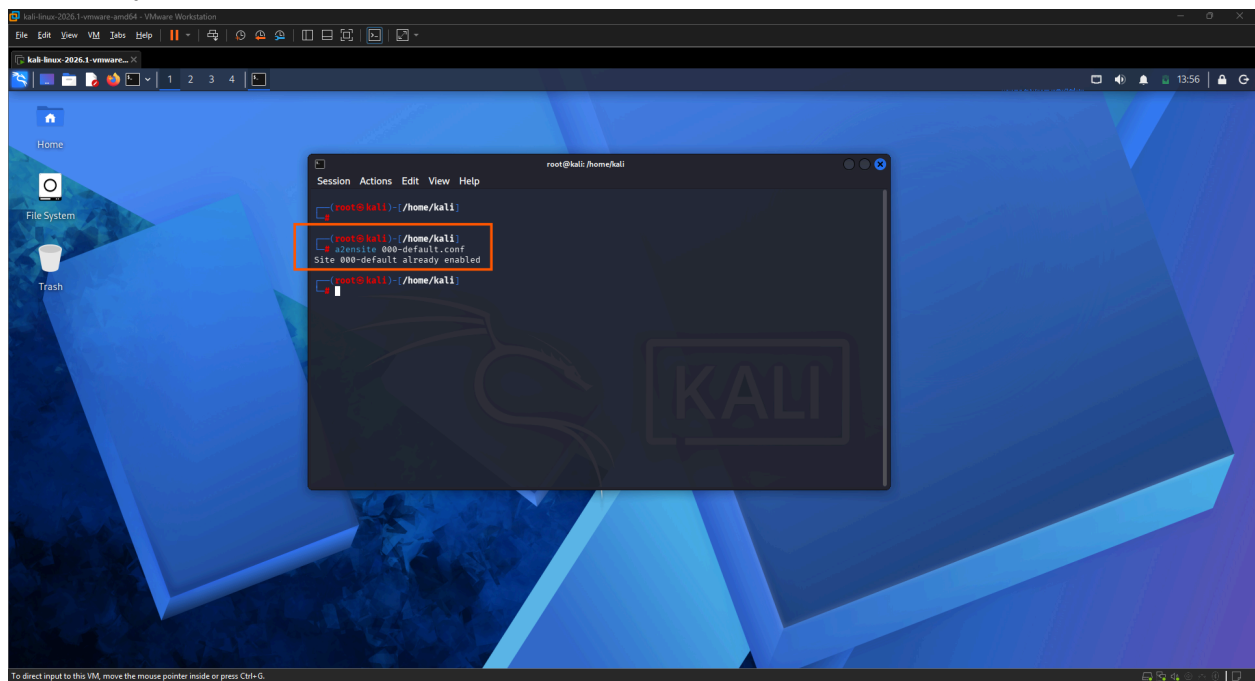
`echo "<html><body><h1>Welcome to mydomain.local</h1></body></html>" | sudo tee /var/www/html/index.html`

Note: This command creates or overwrites `/var/www/html/index.html` with a basic HTML page displaying "Welcome to mydomain.local".

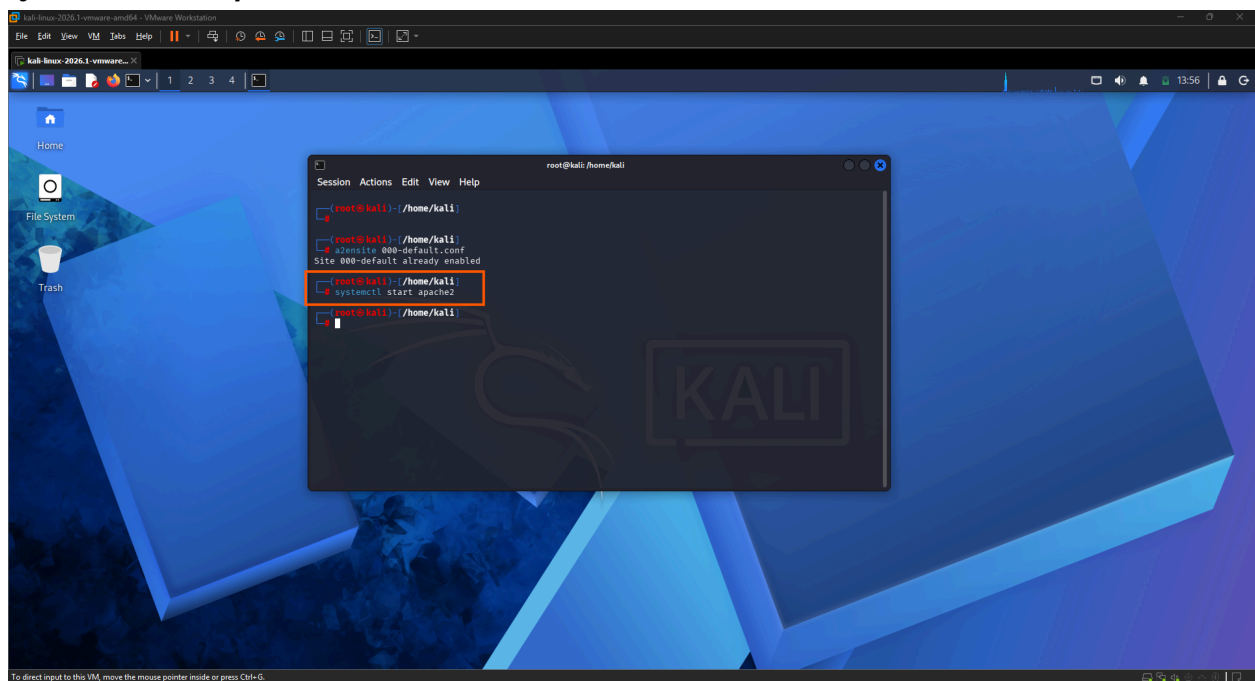


1.8 Use the command: ***a2ensite 000-default.conf***

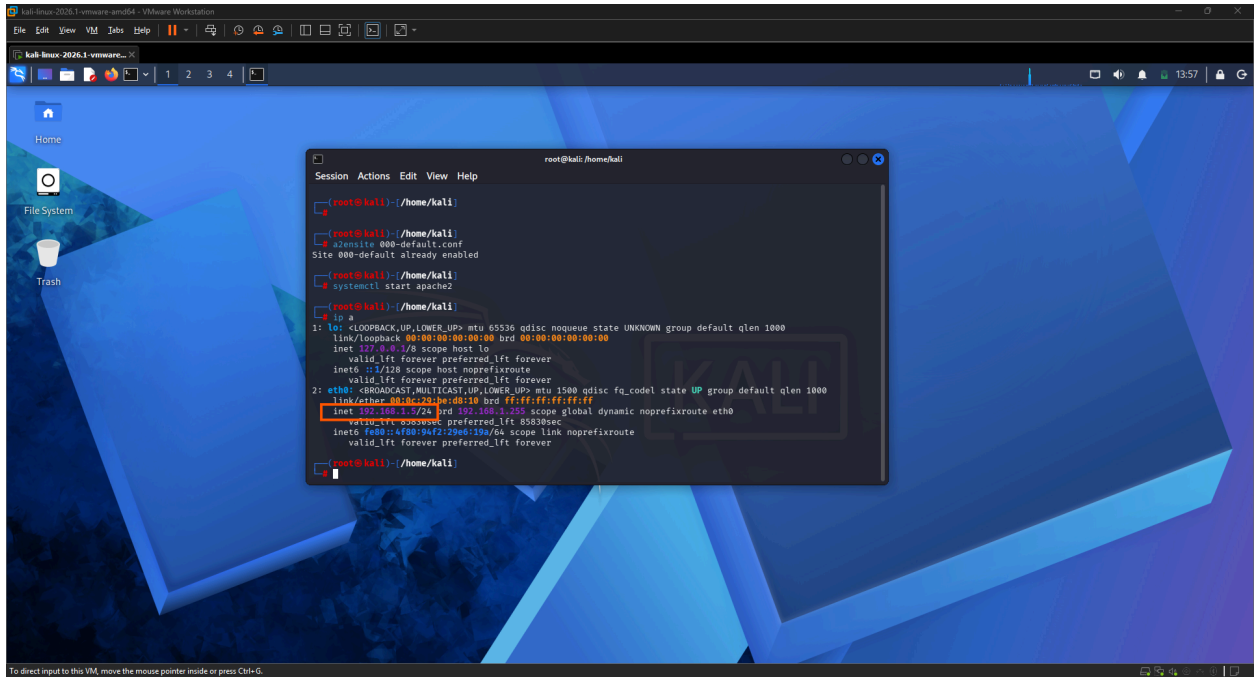
Note: This command enables the Apache site configuration file ***000-default.conf***, allowing it to be used by the web server.



1.9 Start the ***apache2*** using the command:
systemctl start apache2

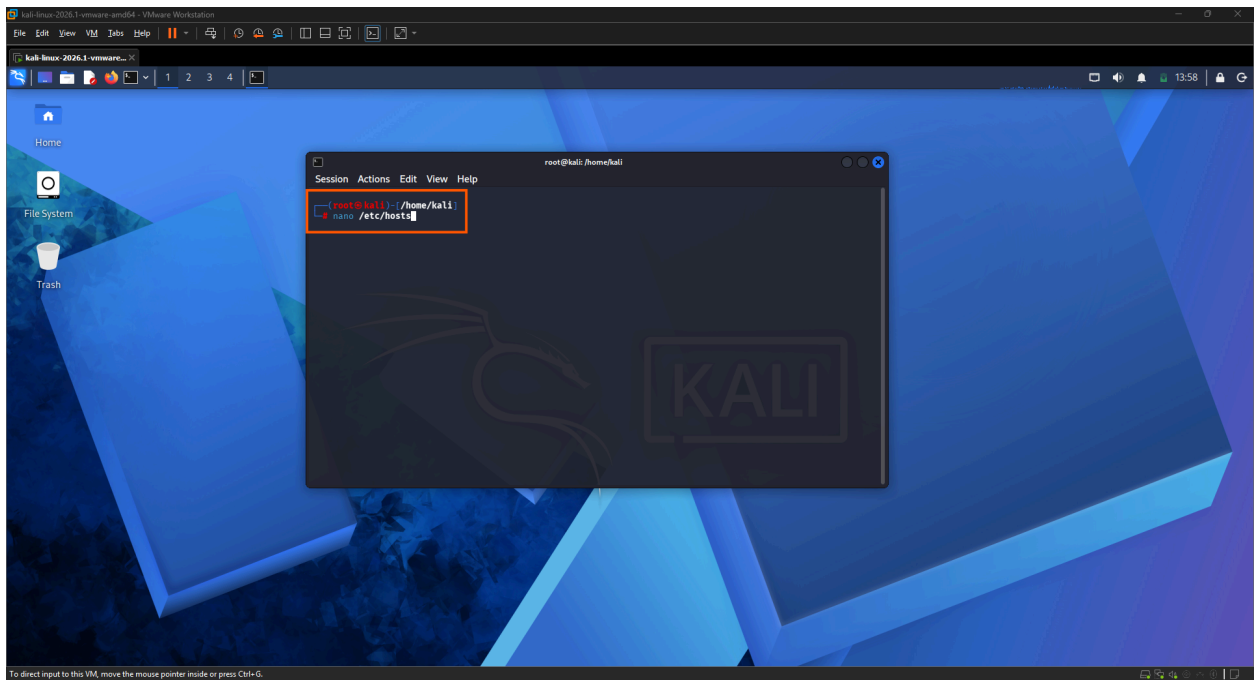


1.10 To see the IP of Kali Linux use the command: ***ip a***
Note: The IP of kali is ***192.168.1.5***



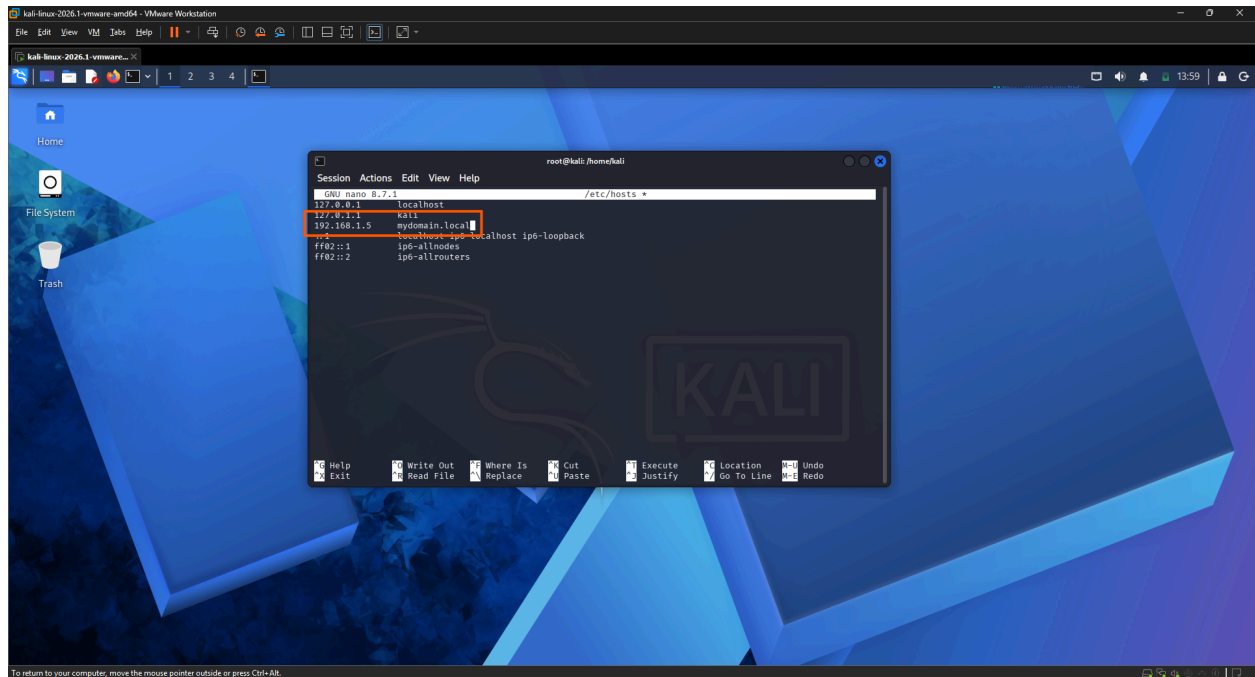
```
root@kali: /home/kali
root@kali: /home/kali
# systemctl start apache2
root@kali: /home/kali
ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:14:00:00 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.5/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
        valid_lft 85838sec preferred_lft 85838sec
    inet6 fe80::4000:5422:20e1:1574 scope link noprefixroute
        valid_lft forever preferred_lft forever
root@kali: /home/kali
```

1.11 Now use the command: ***nano /etc/hosts***

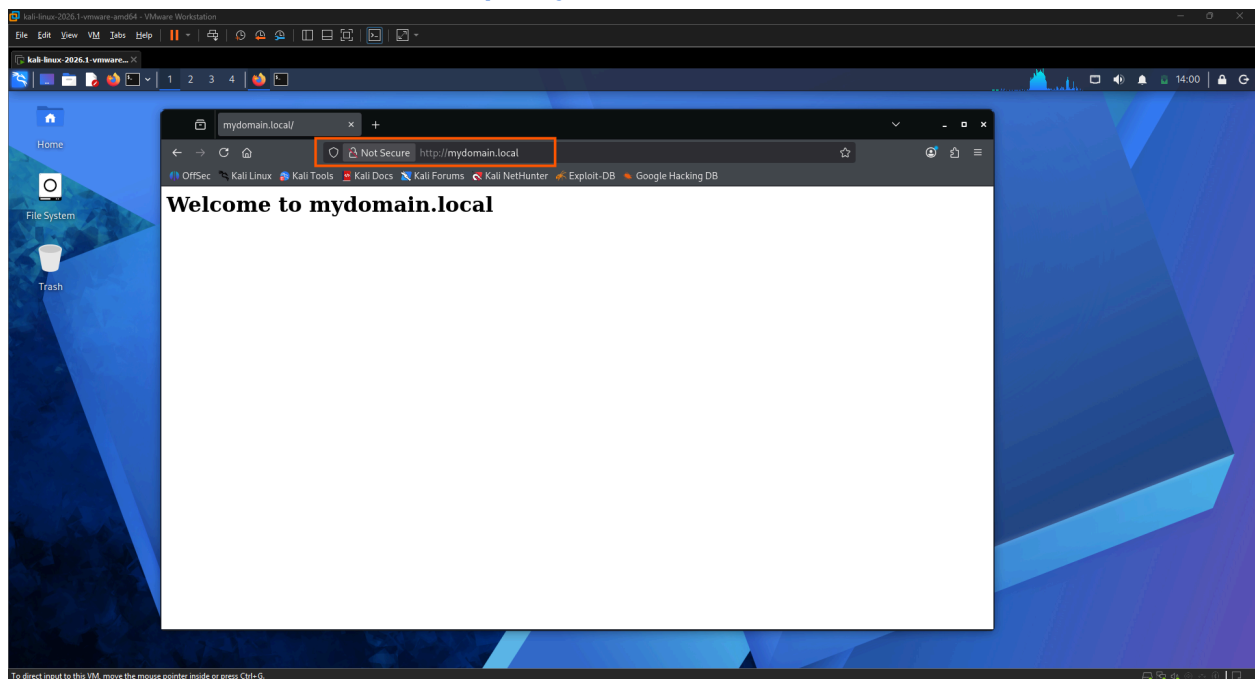


```
root@kali: /home/kali
nano /etc/hosts
```

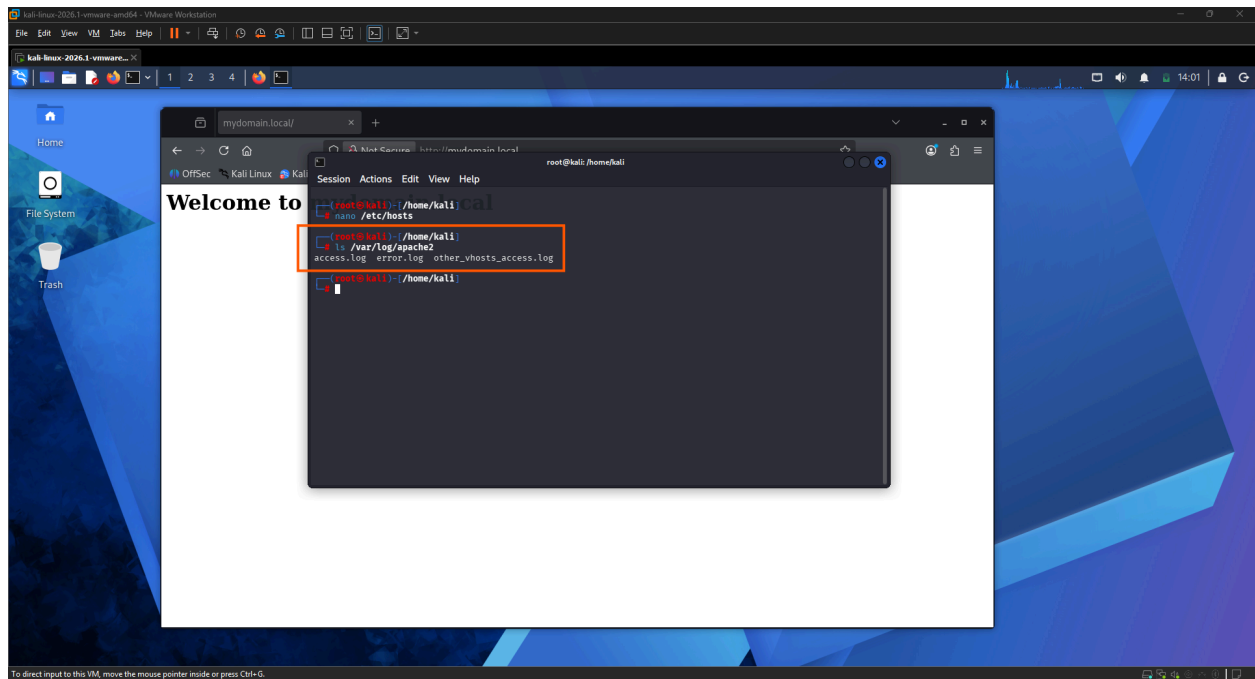
1.12 Add below entry & save the file
192.168.1.5 mydomain.local



1.13 Open the Firefox and search <http://mydomain.local>

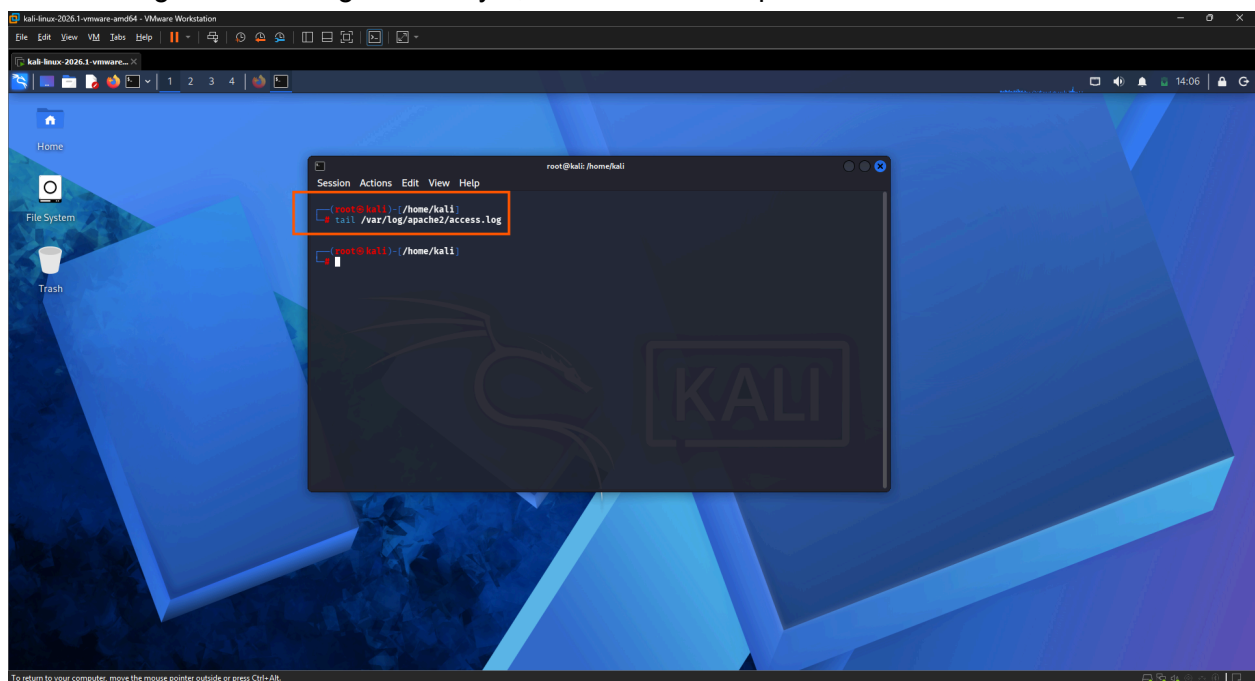


1.14 Now use the command: ***ls /var/log/apache2***



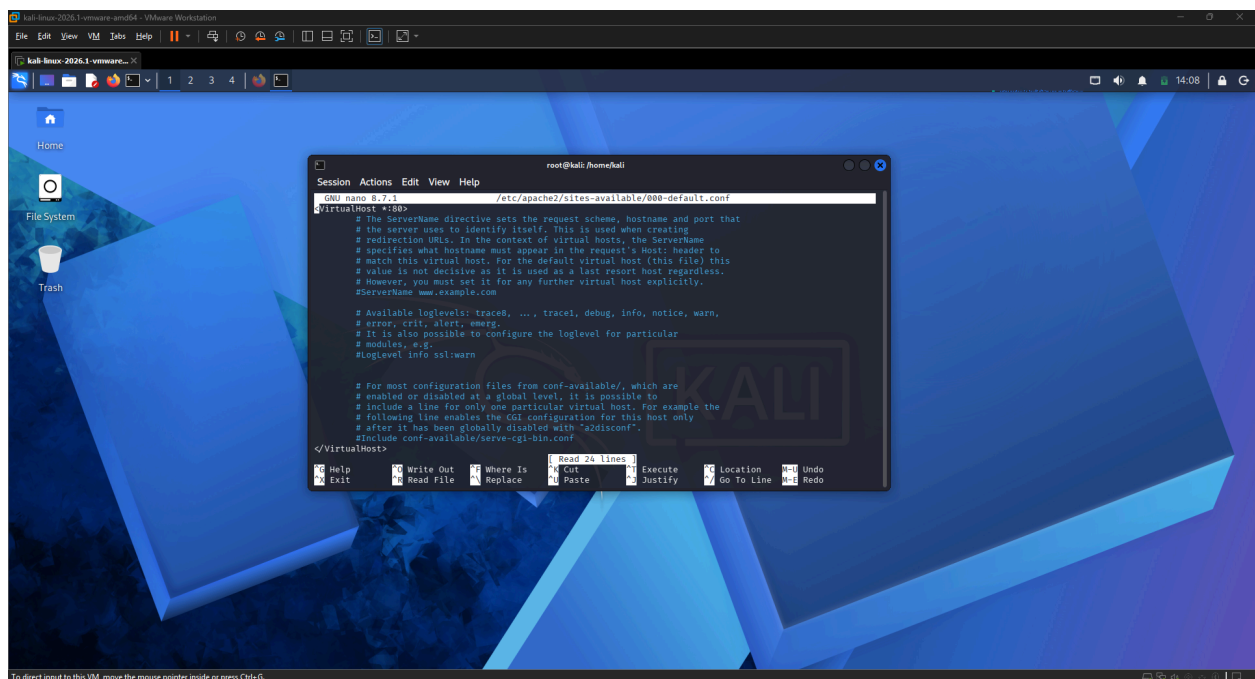
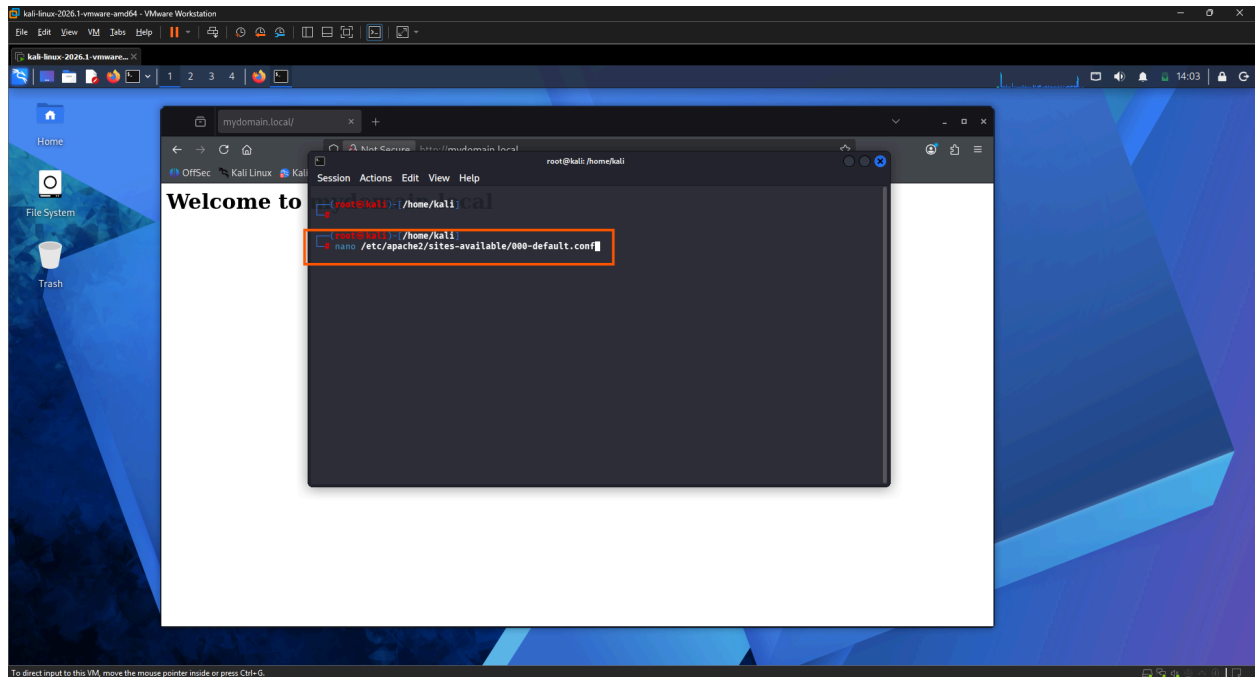
1.15 Use the command: ***tail /var/log/apache2/access.log***

Note: No logs have been generated yet, so there is no output.



Step 2: Adding the entries to log the connections happening to Apache server

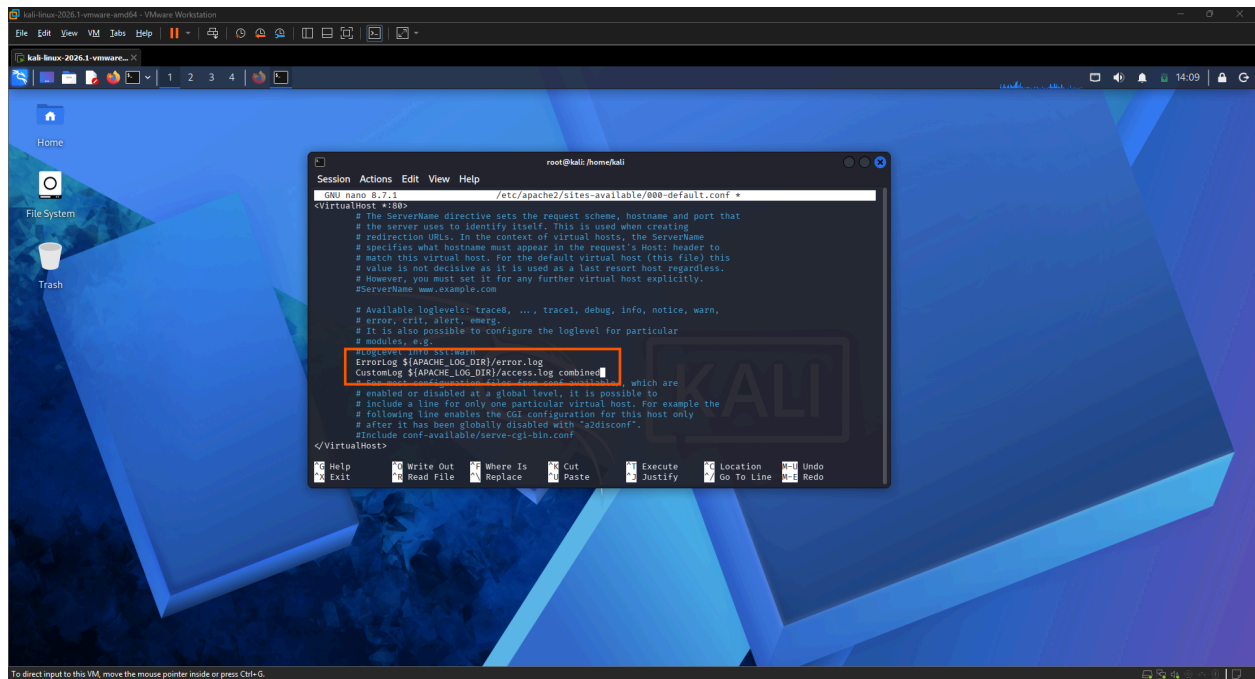
2.1 Now use the command: *nano /etc/apache2/sites-available/000-default.conf*



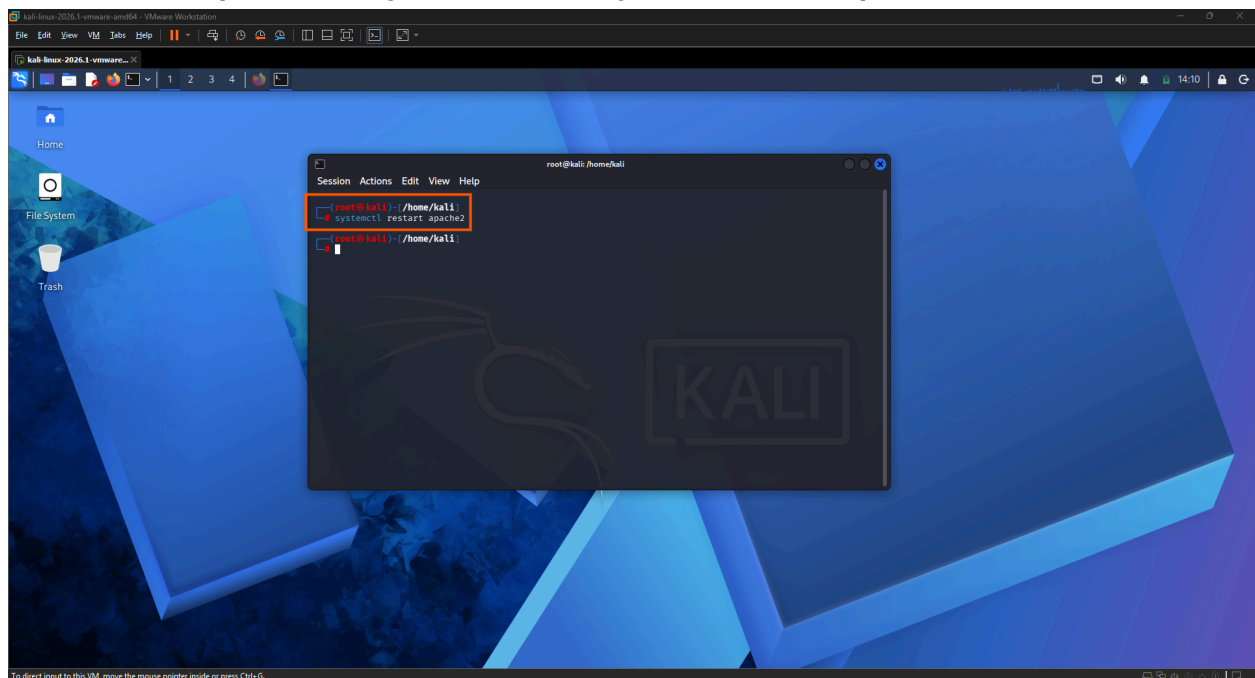
2.2 Append the lines before `</Virtualhost>` & save the file:

ErrorLog `${APACHE_LOG_DIR}/error.log`

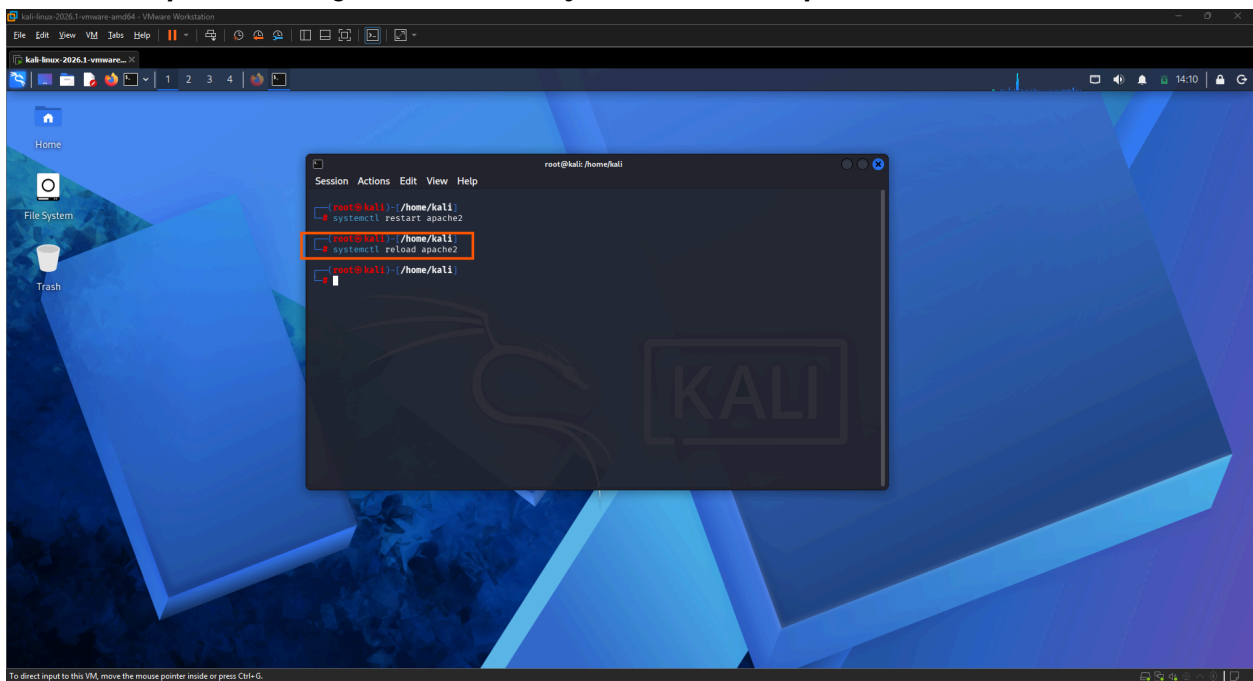
CustomLog `${APACHE_LOG_DIR}/access.log combined`



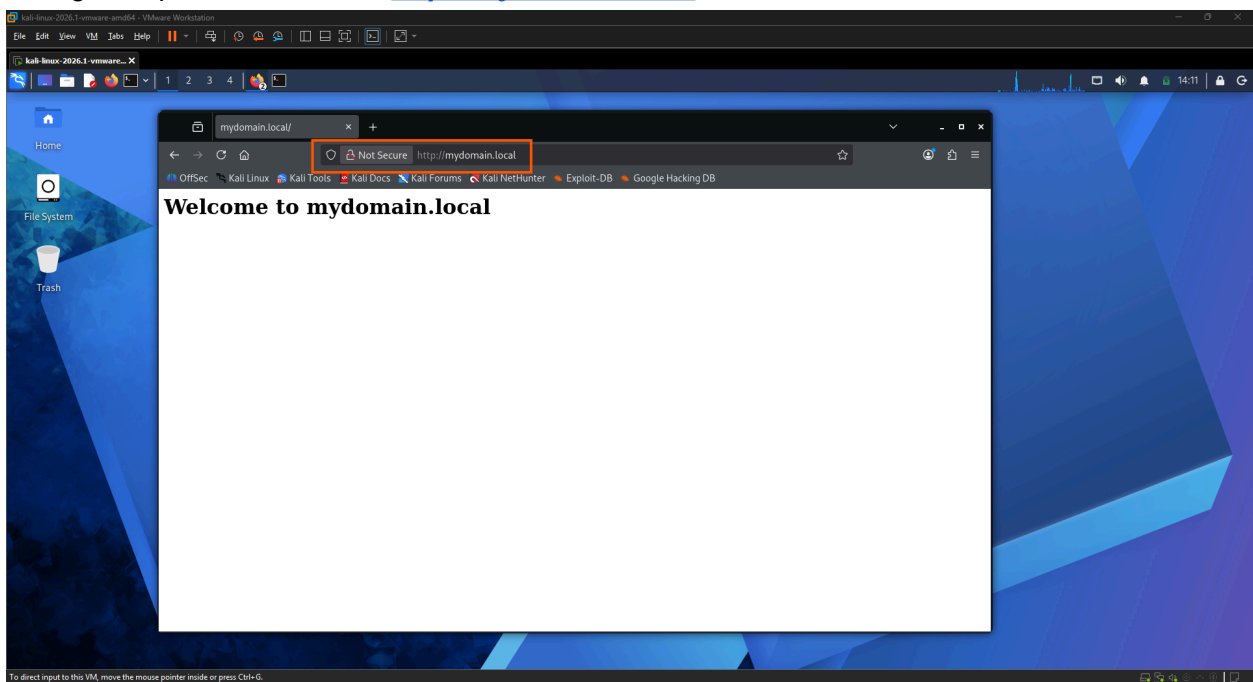
2.3 Restart the **apache2** using the command: **systemctl restart apache2**



2.4 Reload **apache2** using the command: **systemctl reload apache2**

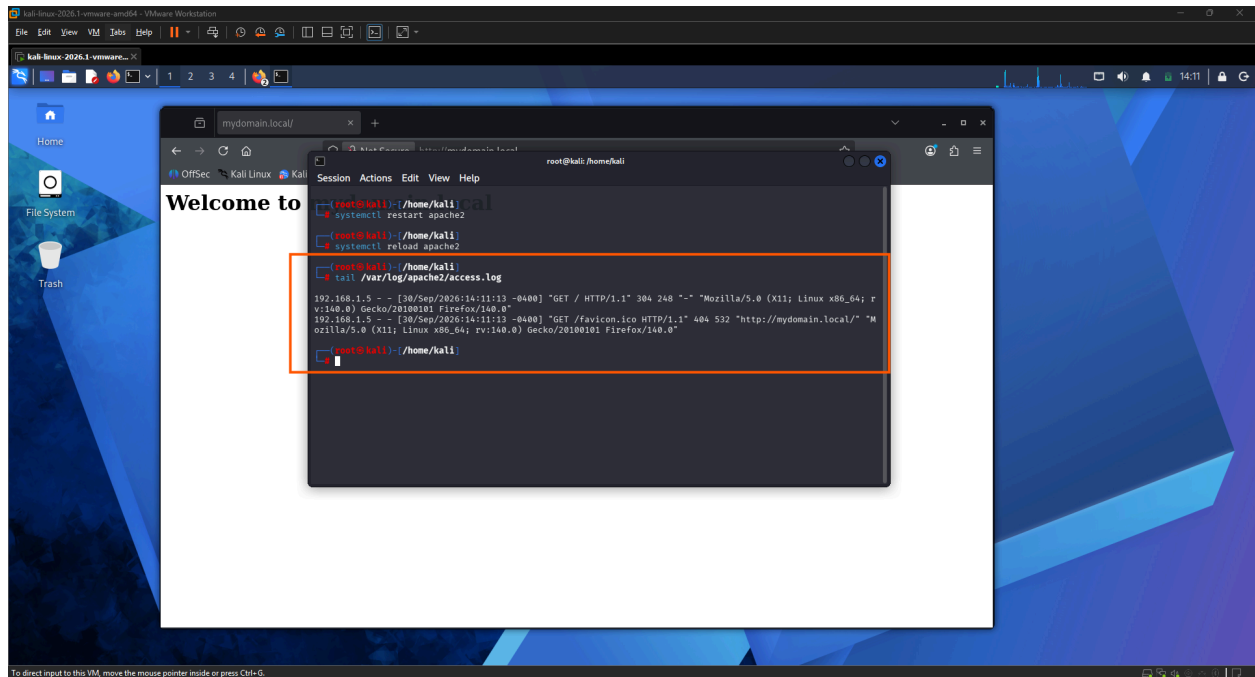


2.5 Again, open in the firefox: <http://mydomain.local>



2.6 Use the command: **`tail /var/log/apache2/access.log`**

Note: Now we can clearly see the logs.



The screenshot shows a Kali Linux desktop environment with a terminal window open. The terminal displays the following commands and output:

```
root@kali: /home/kali
systemctl restart apache2
systemctl reload apache2
tail /var/log/apache2/access.log
```

The output of the `tail` command is highlighted with an orange box and shows the following log entries:

```
192.168.1.5 - - [30/Sep/2026:14:11:13 -0400] "GET / HTTP/1.1" 304 248 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
192.168.1.5 - - [30/Sep/2026:14:11:13 -0400] "GET /favicon.ico HTTP/1.1" 404 532 "http://mydomain.local/" "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0"
```

Conclusion:

The lack of sufficient logging hinders effective monitoring, making it challenging to detect and respond to potential security incidents or system issues. Implementing comprehensive logging is essential to ensure robust monitoring and timely incident management.